

DB Programming

Lab 08 – Thread Management

Lab Objective:

- ✓ Use thread operations, including thread creation and synchronization(joins)
- ✓ Implement thread functions.

➤ Task 1: Creating and destroying threads

When a multi-threaded program starts executing, it has one thread running, which executes the main() function of the program. The program should use the pthread_create() function to create a new thread. Here is how to use it:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h> /* pthread functions and data structures */

/* function to be executed by the new thread */
void* PrintHello(void* data){
    int *my_data = (int*)data; /* data received by thread */
    printf("Hello from new thread : My Data = %d\n", *my_data);
}

/* like any C program, program's execution begins in main */
int main(int argc, char* argv[]){
    int rc; /* return value */
    pthread_t thread_id; /* thread's ID (just an integer) */
    int myData = 1234; /* data passed to the new thread */

    /* create a new thread that will execute 'PrintHello' */
    rc = pthread_create(&thread_id, NULL, PrintHello, &myData);
    if(rc) /* could not create thread */
    {
        printf("\n ERROR: return code from pthread_create is %d \n", rc);
        exit(1);
    }

    printf("\n Created new thread (%d) ... \n", (int)thread_id);
    pthread_exit(NULL); /* terminate the thread */
}
```

1. Type, compile, and run the above code.
2. A thread can get its thread ID by calling `pthread_self()`, which returns the thread ID. We can use it as follows:

```
pthread_t tid;  
tid = pthread_self();
```

Modify the code above to print out the thread ID for the thread in the PrintHello function. Recompile and run the code.

3. Modify the code to create 5 threads, each of which prints out a hello message and its thread ID

➤ Task 2: More on threads

Write a C program that creates an array of integers and an array of threads, both of size 4. Then fill the integer array with the numbers of threads 1,2,3, and 4. Then, use a different thread to add 10 to the values of the integer array. You need to print the resulting array on the screen.

Note the following:

1. You need a loop to create multiple threads.
2. You need a loop to join all threads back in the main.

The End